

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | Kent, OH - Remote (New York metro)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) building production RAG systems, LLM evaluation pipelines, and agentic workflows - the exact infrastructure Arize Phoenix and Arize AX are designed to monitor. Daily practitioner of LLM tracing, hallucination measurement, retrieval quality eval, and agent orchestration debugging in healthcare AI contexts. Track record of shipped technical systems: ~35% retrieval precision improvement, >30% hallucination reduction, production FastAPI/Docker services with structured audit logging. Brings a practitioner's perspective to developer relations: can write tutorials developers actually run because I've solved these problems hands-on.

CORE COMPETENCIES

LLM Evaluation & Observability • RAG Systems & Retrieval Quality • LangChain / LlamaIndex • Agentic Workflow Debugging • Hallucination Measurement • Guardrails & Grounding • Technical Writing & Documentation • OSS Project Delivery • Developer Demo Design • Python / FastAPI • TypeScript / React • Docker / CI/CD • Vector Search / Embeddings • Prompt Engineering

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare AI
USA

- Built end-to-end **LLM evaluation pipelines** with grounding rules, guardrails, hallucination tracking, and audit trails for production healthcare AI - the same observability layer Arize Phoenix provides as a platform.
- Designed **RAG systems** for clinical knowledge retrieval: recursive semantic chunking, transformer-based embeddings, vector search, retrieval quality eval loop; precision improved ~35%; retrieval-grounded alignment exceeded **90%**.
- Built **agentic LLM workflows** (LangChain) for multi-step healthcare queries - structured outputs, tool discipline, intermediate trace logging; reduced hallucinations >30%, improved agent stability ~25%.
- Shipped **FastAPI/Docker** inference services with structured logging and load simulation - reduced post-deployment defects ~30%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech
Hyderabad, India

- Implemented **document Q&A with RAG** - reduced hallucinated outputs ~30%; deployed as Flask API adopted independently by non-technical staff.
- Built transformer-based **video summarization** system - reduced review time **60-70%**; ~85% **alignment** with human-curated highlights.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)
Hyderabad, India

- Built **Jenkins CI/CD pipelines** and **Grafana monitoring dashboards** for SQL deployments - reduced errors >30%, improved release cycles ~35-40%.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline with **schema-validated JSON contracts between agents**, RAG-grounded CPT/ICD validation, ANN duplicate detection, and explainable audit-ready reasoning traces - a complete example of the agentic observability challenges Arize Phoenix addresses.

LangChain, LlamaIndex, Python, FastAPI, Docker, vector search

Manga Lens - Chrome Extension for Real-Time AI Translation

Shipped independently to Chrome Web Store. Full-stack browser extension integrating **four AI providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local) - multi-provider tracing and payload management at scale.

TypeScript, Manifest V3, Python, multi-provider LLM integration

Dream Decoder - AI-Powered Multimodal Creative Platform

FastAPI + React/TypeScript app with structured prompt transformation layers; improved contextual alignment **~30%** over direct-prompt orchestration.

FastAPI, React, TypeScript, Perplexity Sonar, Replicate

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

LLM Observability & Evals: LLM tracing, hallucination measurement, retrieval quality evaluation, guardrails, grounding, audit logging, structured outputs

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, prompt engineering, embeddings, vector search

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL (PostgreSQL, T-SQL)

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, YOLOv8

Infra: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs

Work Authorization: F-1 OPT - authorized to work in the United States now; future sponsorship required